PROJECT 2

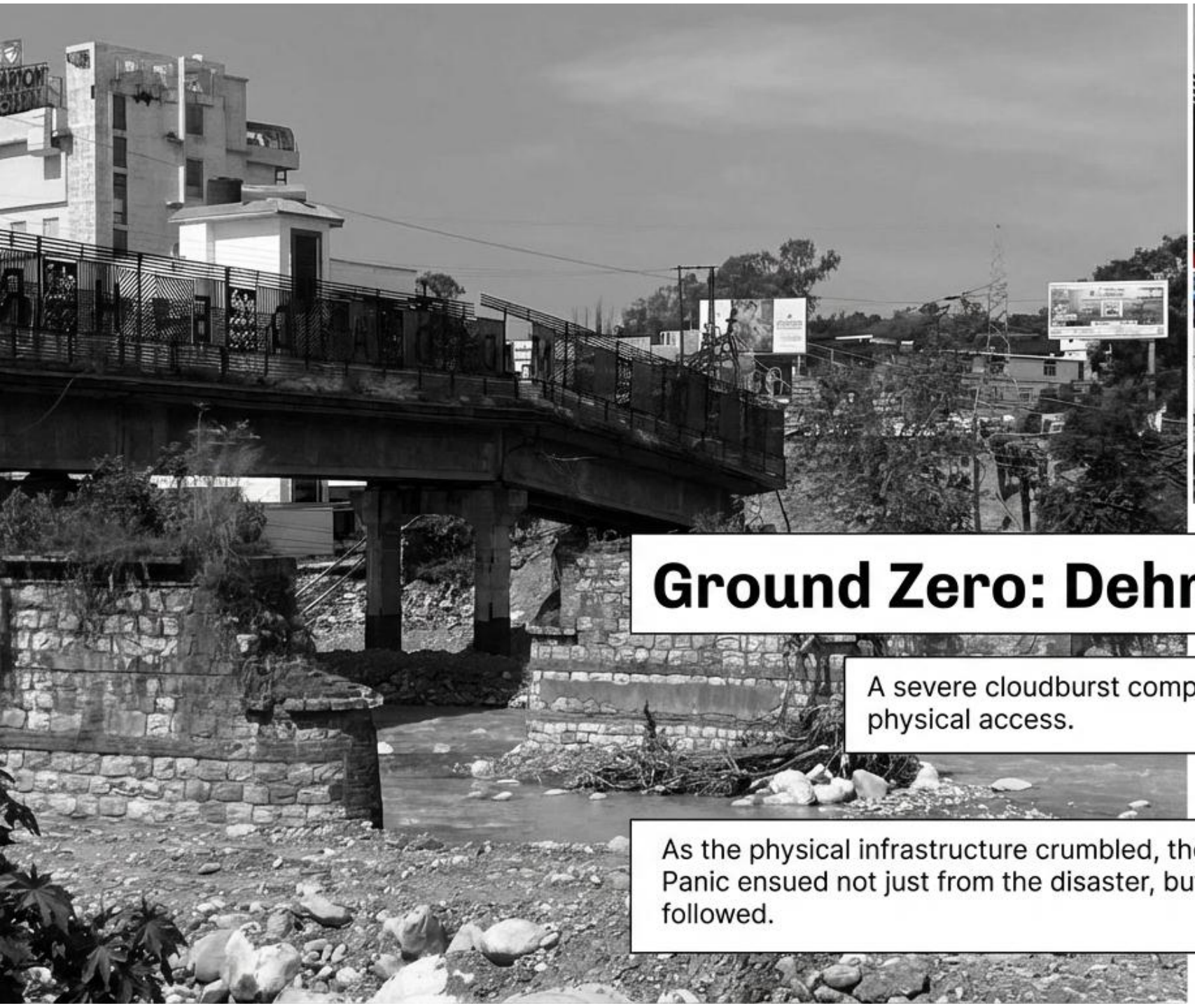


Emergency Communication Systems (EBS)

A Low-Tech Solution for High-Stakes Environments

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Mechanical Engineer
Transportation designer



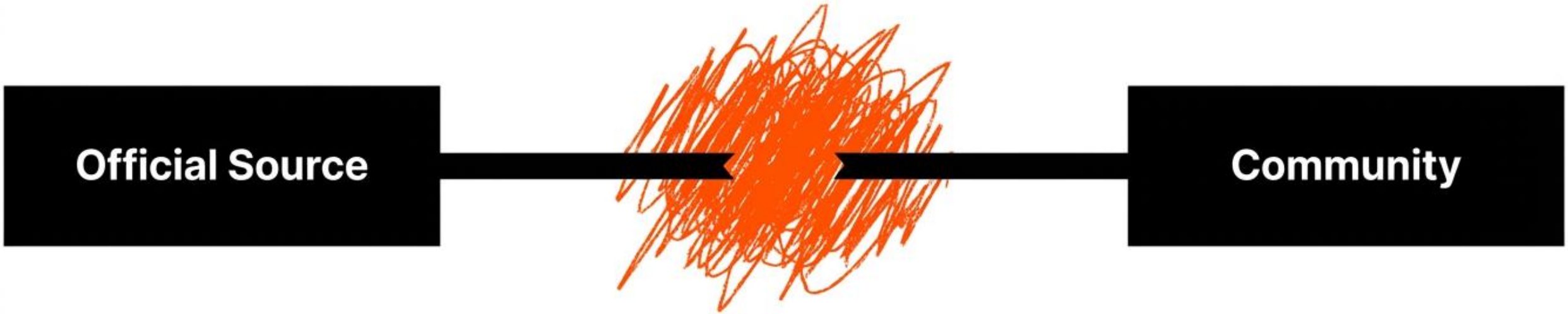


Ground Zero: Dehradun Cloudburst

A severe cloudburst completely collapsed a vital bridge, severing physical access.

As the physical infrastructure crumbled, the communication infrastructure followed. Panic ensued not just from the disaster, but from the sudden, terrifying silence that followed.

The Communication Void



When official channels fail, rumor and panic spread faster than facts.

The traditional system's inability to deliver verified, accessible updates forces communities to rely on unverified noise. This project focuses on building trust and breaking through that noise.

Diagnostic Matrix: Infrastructure Failure

Word-of-Mouth	Traditional Media (TV & Radio)	Official Alert Systems	Networked Cell Towers
○			✕
Relying on public gossip and old newspapers is highly unreliable and fundamentally slow to adapt.	Broad and generic. Fails instantly when residential electricity drops. Cannot provide provide hyper-local, immediate updates.	Centralized websites lock up when internet dies; local sirens are hopelessly dependent on the fragile power grid.	Fiber optics and cell towers require constant power and physical safety. In a disaster, the network clogs instantly and collapses.

Mixed-Methods On-Ground Research

The Officer

"Received deployment orders from senior officials. Admitted there is no dedicated disaster department for informing the public."

The Security Guard

"First learned about the collapse via a video sent on WhatsApp. Highly skeptical of the video and the news."

The Shop Vendor

"Got the news not from official sources, but from a WhatsApp video. There is no fixed order to get updates."

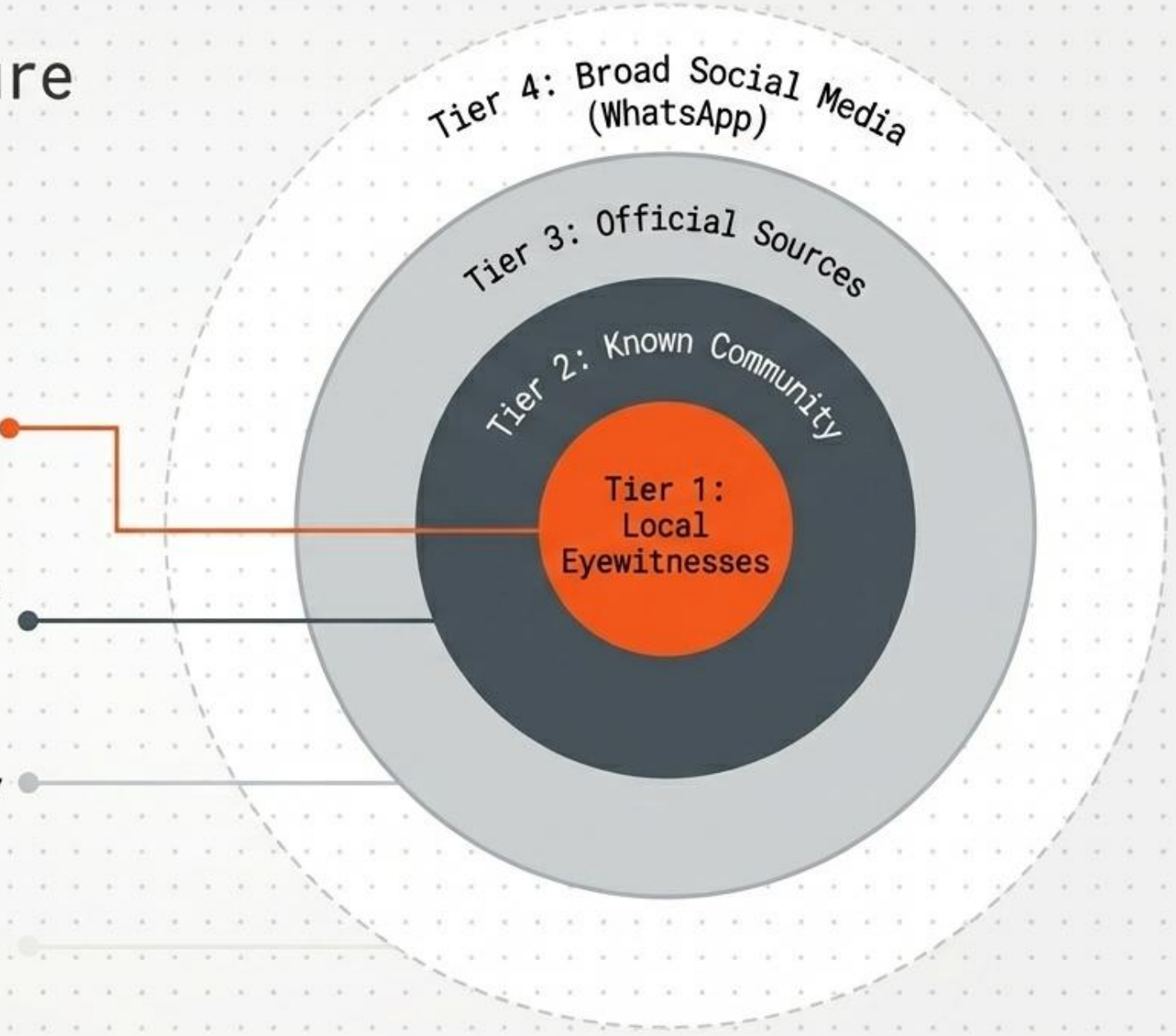
The Trust Architecture

The highest degree of trust. Reliable information comes from local people who saw the incident firsthand.

High trust. People trust information from known contacts and community members over large-scale sources.

Tier 3: Official Sources
Delayed. Often randomized in delivery and reliant on functioning internet.

Tier 4: Broad Social Media (WhatsApp)
High skepticism. Often the fastest to arrive but viewed with intense suspicion as public gossip.



The Stakeholder Ecosystem



Travellers

Require dynamic, routing-based digital interventions. (Solution path: Redesigning Govt. Mausam App).



Rural Areas (Remote Villages)

Require off-grid, physically resilient warning systems. (Solution path: EBS Hardware Device).

Core Goal: Tackle disaster silence by building a trusted, two-way communication channel accessible even when traditional networks fail.

The Hardware Design Brief

Constraint 1: Grid Independence

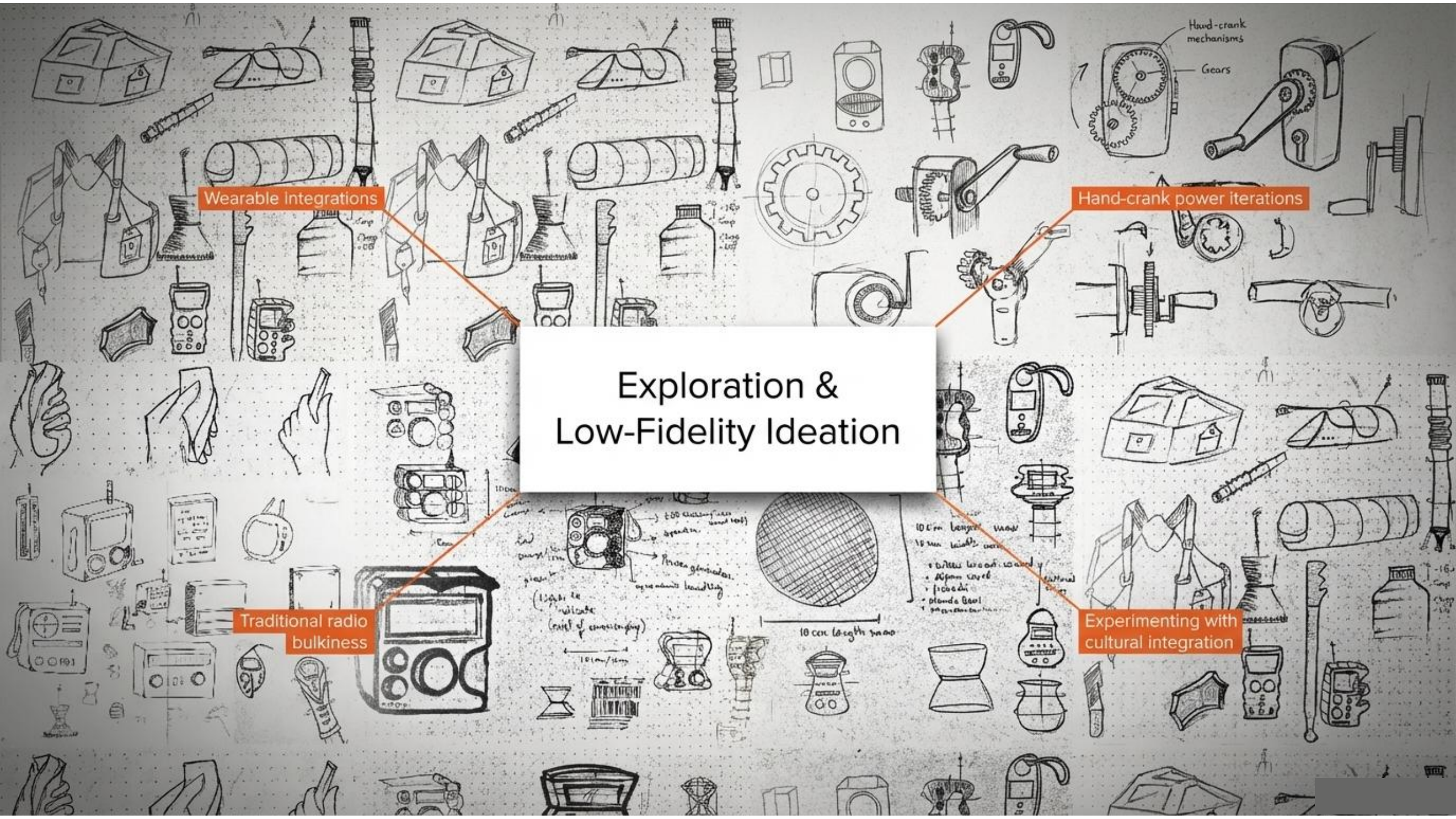
Must operate completely off the power and cellular grid using low-tech, durable components.

Constraint 2: Verified LoRaWAN

Must utilize a LoRaWAN mesh network to deliver authenticated, hyper-local warnings to remote areas.

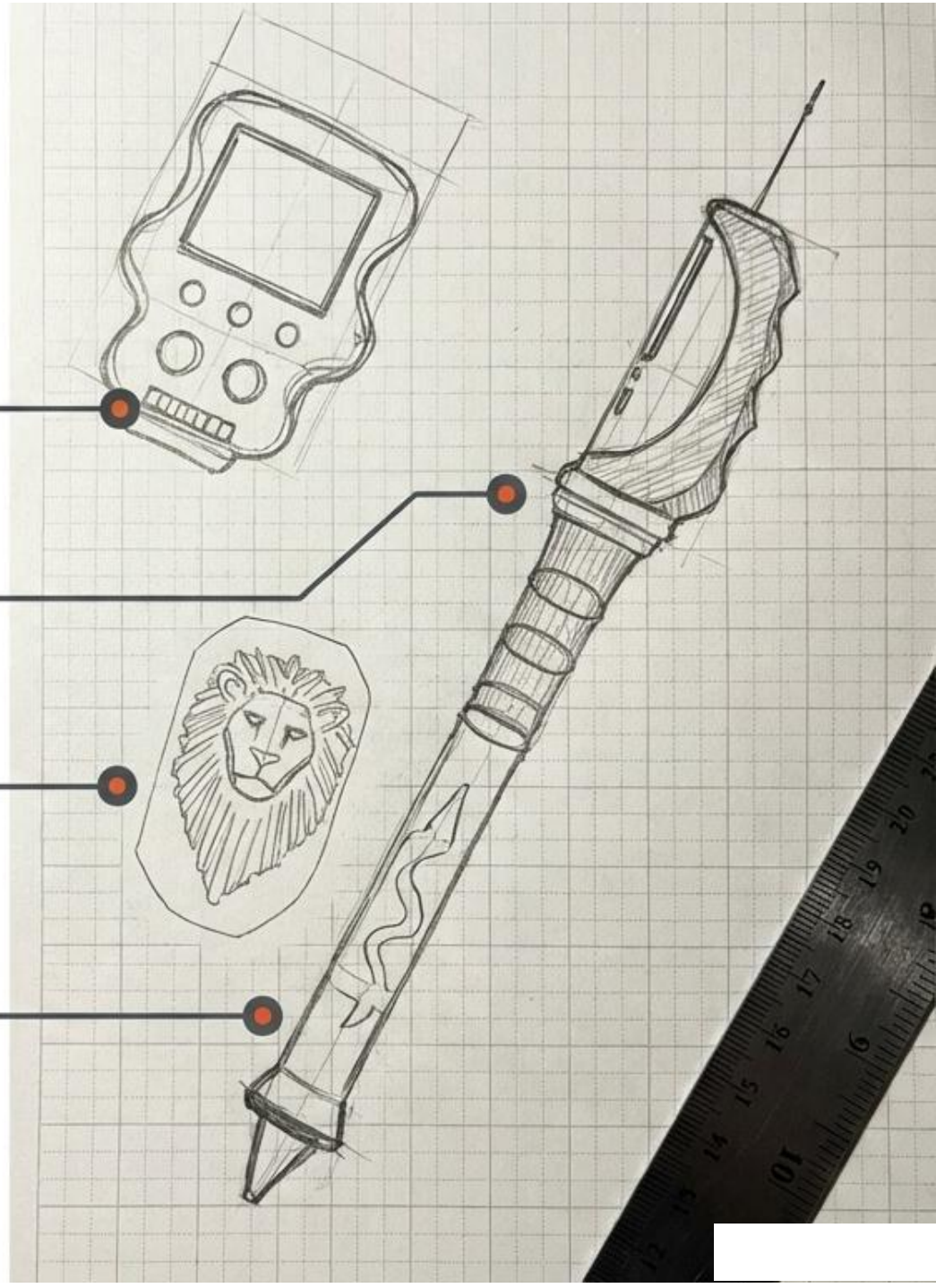
Constraint 3: Muscle Memory UX

Must replace cognitive overload and panic with automatic, muscle-memory driven action through intuitive physical dials and buttons.


Hand-crank mechanisms
Gears
Wearable Integrations
Hand-crank power iterations
Exploration & Low-Fidelity Ideation
Traditional radio bulkiness
Experimenting with cultural integration

Concept Refinement: The Trekking Pole Paradigm

Contextual Form Factor: Transitioning away from bulky radios to a form factor native to the Himalayan/Dehradun terrain.



Hardware Anatomy: EBS Guardian



Channel Dial

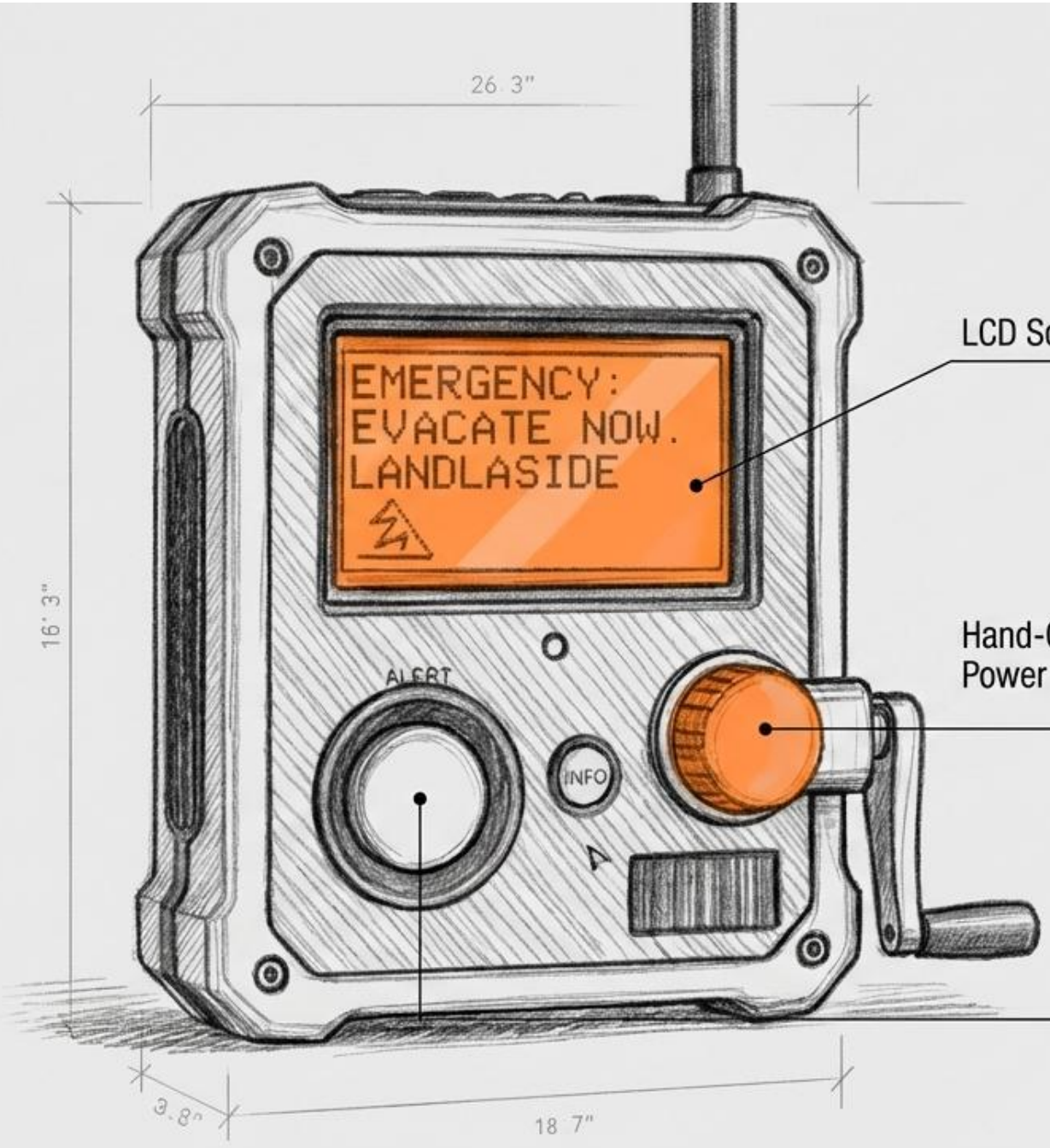
LoRaWAN Antenna

Detachable Locking Mechanism

Ergonomic Hand Grip

LED Alert Light

Telescopic Aluminium Staff Segment



LCD Screen

Hand-Crank
Power Generator

Interface UX: Detachable Handheld I/O

LCD Screen

Low-resolution, high-contrast. Forces brevity. Displays stark text and icons: 'EMERGENCY: EVACUATE NOW. LANDSLIDE'.

Hand-Crank Power Generator

Ensures infinite off-grid operation via manual kinetic energy.

Physical Controls

Large, tactile Emergency Alert Button and status LED indicator. Designed for operation with compromised fine motor skills.